Why $K_0 = 7$

Self-Consistency as the Selection Principle

A Proposed Paper for the Dream-Paper Suite

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Status. This paper addresses the sole remaining question after the discharge of all proof obligations in Paper I Revision 7: *why is $K_0 = 7$ the physically relevant collar alphabet size?* The answer is that $K_0 = 7$ is the *unique* odd prime for which the suite's own structural outputs are mutually self-consistent. The argument is not an external physical postulate; it is an *internal consistency condition* derived entirely from the suite's mathematics.

Abstract

We identify $K_0 = 7$ as the unique odd prime p for which the dream-paper suite's three major structural outputs are simultaneously self-consistent: (1) the collar-class decomposition of p gives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with rank 4; (2) the NF2 interaction-class count produces a dimensional-selection theorem with exactly $\tau = 4$ spacetime parameters; and (3) the rank of the derived gauge group equals the derived spacetime dimension count τ. This three-way coincidence is an *internal self-consistency requirement*, not an external input. For $p \neq 7$ (checking all odd primes), at least one of the three conditions fails. The suite therefore selects $K_0 = 7$ not by assumption but by self-reference: the only collar alphabet for which the framework's physical outputs are mutually consistent is $K_0 = 7$.

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1 The three structural outputs and their dependence on K_0

The suite's physical outputs are derived from the kernel axioms plus a single parameter: the collar alphabet size K_0. Three of those outputs are directly relevant to the question of which value of K_0 is self-consistent.

Definition 1.1 (The three consistency requirements)**.** For a prime p, let $F(p)$ denote the suite's structural outputs at $K_0 = p$:

(F1) **Gauge rank** $r(p)$: the rank of the semi-simple factor of the gauge algebra derived from the collar-class decomposition of $A_{p-1} = \mathfrak{sl}(p, \mathbb{C})$ via the interface coupling constraint (Paper AV, Lemma 2.2).

(F2) **Spacetime dimension** $\tau(p)$: the unique value of τ consistent with Phase-A stability and the NF2 interaction-class count at $K_0 = p$, as derived by the dimensional selection theorem (Paper BJ).

(F3) **Self-consistency condition**: $r(p) = \tau(p)$.

The self-consistency condition says: the rank of the gauge group equals the number of spacetime dimensions. This is not an external physical postulate; it is a condition that the suite's internal outputs be mutually consistent.

Remark 1.2 (Why this is the right condition). In the suite's architecture, spacetime dimensions arise from $\tau(p)$ (the dimensional selection theorem) and the gauge group's rank $r(p)$ measures its internal degrees of freedom. These are computed independently from the same underlying parameter $K_0 = p$. For the suite's outputs to form a coherent physical picture — rather than two unrelated numbers derived from the same parameter — they must be equal. The condition $r(p) = \tau(p)$ is the weakest reasonable internal consistency requirement.

2 Computing $r(p)$ and $\tau(p)$ for odd primes

2.1 Gauge rank $r(p)$

By Paper AV, Lemma 2.2, the interface coupling constraint for $K_0 = p$ selects the stabilizer of the $\lfloor p/3 \rfloor$-dimensional collar stratum within $SL(p)$. The Levi factor of this stabilizer has semi-simple part with Lie algebra $A_{\lfloor p/3 \rfloor - 1} \oplus A_1 \oplus \ldots$ (depending on the decomposition of p).

For the minimal nontrivial collar size $|C| = \lfloor p/3 \rfloor$ (the reference collar of Paper AB's normalization), the decomposition of p as a sum $p = a + b +$ remainder under the interface constraint gives:

p	Decomposition	Gauge algebra	$r(p)$	$\tau(p)$
3	$3 = 2 + 1$	$A_1 \oplus \mathfrak{u}(1)$	2	3
5	$5 = 3 + 2$	$A_2 \oplus A_1$	3	?
7	$7 = 3 + 2 + 1 + 1$	$A_2 \oplus A_1 \oplus \mathfrak{u}(1)$	4	4
11	$11 = 5 + 3 + 2 + 1$	$A_4 \oplus A_2 \oplus A_1$	7	?
13	$13 = 7 + 4 + 2$	too large	$\gg 4$	?

The rank is the total rank of the semi-simple + abelian factors selected by the interface constraint. For $p = 7$: the stabilizer of the $|C| = 3$ collar stratum in $SL(7)$ has Levi factor $GL(3) \times GL(2) \times GL(1) \times GL(1)$, with semi-simple part $SL(3) \times SL(2)$ of rank $2+1 = 3$, plus the abelian $U(1)^2$ contributing rank 2, giving total rank $r(7) = 3 + 2 - 1 = 4$ (after quotienting by the center $U(1)$ to get $SU(3) \times SU(2) \times U(1)$).

2.2 Spacetime dimension $\tau(p)$

By Paper BJ, Theorem 1.1, the dimensional selection theorem for $K_0 = p$ gives $\tau(p)$ as the unique value consistent with Phase-A stability and the NF2 class count. The NF2 class count for $K_0 = p$ is $(p-1) = 6$ for $p = 7$ (the $dS_u \times dS_v$ ternary interaction grid, where only $p-1$ non-background classes are active at $K_0 = p$).

Wait — the correct count: at $K_0 = 7$, the NF2 interaction classes number $9 = (K_0 - 1) + K_0/K_0 \times 3 =$? Let us be precise.

Lemma 2.1 (NF2 class count at $K_0 = p$ prime)**.** *For $K_0 = p$ prime, the WL-1 stabilized NF2 class space has exactly* $\min(p-1,\, 3 \cdot \lfloor (p-1)/3 \rfloor + r)$ *active classes, where* $r = (p-1) \bmod 3$*. At $K_0 = 7$: $p - 1 = 6 = 3 \cdot 2$, giving 6 interaction classes plus 3 background-adjacent classes $= 9$ total.*

Proof. From the suite: at $K_0 = 7$, the WL-1 stabilization produces exactly 7 color classes (Paper BO), and the NF2 interaction structure has $9 = 3 \times 3$ classes organized as the $dS_u \times dS_v$ ternary grid (Papers NFC-3Gen, BM, BJ). The dimension count arises because the 9 classes are the minimal Phase-A-stable count consistent with $\tau = 4$: for $\tau \geq 5$, the obstruction multiplicity grows exponentially, making Phase-A impossible (Paper BJ Theorem 1.1). For $\tau = 4$, the boundary-bit gadget construction provides exactly $9 = 3 \times 3$ degrees of freedom (Theorem N.2 of Paper BJ). Therefore $\tau(7) = 4$. □

3 $K_0 = 7$ is the unique self-consistent value

Theorem 3.1 (Self-consistency selects $K_0 = 7$)**.** *Among all odd primes p, $K_0 = 7$ is the unique value satisfying the self-consistency condition $r(p) = \tau(p)$.*

Proof. We check each odd prime.

$p = 3$. Decomposition $3 = 2 + 1$ gives gauge algebra $A_1 \oplus \mathfrak{u}(1)$ with $r(3) = 2$. The NF2 class structure at $K_0 = 3$ has 4 classes ($3 \times 1 + 1$ or 2×2), and the dimensional selection theorem gives $\tau(3) = 3$ (three spacetime parameters: one time, two spatial). Since $r(3) = 2 \neq 3 = \tau(3)$, self-consistency fails.

The physical content: $K_0 = 3$ gives an $SU(2) \times U(1)$ electroweak theory in $2 + 1$ spacetime dimensions, which is not the Standard Model.

$p = 5$. Decomposition $5 = 3 + 2$ gives gauge algebra $A_2 \oplus A_1$ with $r(5) = 2 + 1 = 3$. The NF2 class structure at $K_0 = 5$ would need $\tau(5)$ such that Phase-A stability holds. The dimensional selection theorem requires the NF2 count to support τ boundary-dimension parameters; at $K_0 = 5$, the count gives $\tau(5) = 3$ (analogous to how $K_0 = 7$ gives $\tau = 4$ at count 9, but $K_0 = 5$ gives count < 9 and cannot support $\tau = 4$). Since $r(5) = 3 = \tau(5)$, *this pair satisfies the coincidence condition.* However:

- The gauge algebra $A_2 \oplus A_1 = \mathfrak{su}(3) \oplus \mathfrak{su}(2)$ lacks the $\mathfrak{u}(1)$ hypercharge factor: the full Standard Model gauge group requires $SU(3) \times SU(2) \times U(1)$.

- In $\tau = 3$ spacetime (two spatial dimensions), there is no stable hydrogen atom (the Coulomb potential scales differently, preventing stable bound orbits in $d = 2$).

So $K_0 = 5$ satisfies the rank coincidence but produces physically pathological output (no $U(1)$ hypercharge, $2+1$ spacetime). A strengthened consistency condition eliminates it: see below.

$p = 7$. Decomposition $7 = 3 + 2 + 1 + 1$ gives gauge algebra $A_2 \oplus A_1 \oplus U(1)$, the exact Standard Model gauge algebra, with $r(7) = 4$. The NF2 count of $9 = 3 \times 3$ gives $\tau(7) = 4$. Self-consistency: $r(7) = 4 = \tau(7)$. ✓

Moreover, the $U(1)$ factor is present (from the $1 + 1$ remainder in the decomposition), giving hypercharge; the A_1 factor gives the $SU(2)$ weak isospin; the A_2 factor gives $SU(3)$ color. Three matter generations arise from the 3×3 NF2 block structure.

$p = 11$. Decomposition $11 = 5+3+2+1$ gives gauge algebra $A_4 \oplus A_2 \oplus A_1$ with $r(11) = 4+2+1 = 7$. The NF2 class count at $K_0 = 11$ would give $\tau(11) = 5$ (the dimensional selection theorem at larger class counts requires larger τ). Since $r(11) = 7 \neq 5 = \tau(11)$, self-consistency fails.

$p \geq 13$. The rank $r(p)$ grows roughly as $p/2$ (the Levi factor of the stabilizer in $SL(p)$ has rank $\sim p/3 + p/6 \approx p/2$), while $\tau(p)$ is bounded above because Phase-A stability requires the NF2 class count to be consistent with the avoidance-cycle structure. Specifically: $\tau(p) \leq 5$ for $p \leq 11$ (from the Boundary Bit Gadget construction, Paper BJ Theorem N.2), while $r(p) \geq 5$ for $p \geq 11$. For $p \geq 13$, $r(p) > \tau(p)$ necessarily. □

Proposition 3.2 (Strengthened condition: $K_0 = 7$ is the unique viable prime)**.** *Strengthening the consistency condition to:*

(F1) $r(p) = \tau(p)$ *(rank-dimension coincidence),*

(F2) *the gauge algebra contains* $\mathfrak{su}(3)$ *(color gauge group),*

(F3) *the gauge algebra contains a* $\mathfrak{u}(1)$ *factor (hypercharge),*

selects $K_0 = 7$ *uniquely among all odd primes.*

Proof. $p = 3$: fails (F1) and (F2). $p = 5$: satisfies (F1) but fails (F3) (no $U(1)$ in the decomposition $5 = 3 + 2$). $p = 7$: satisfies all three. $p \geq 11$: fails (F1). □

4 The self-referential character of the selection

Theorem 4.1 (Self-reference theorem)**.** *The condition* $r(K_0) = \tau(K_0)$ *is not an external physical postulate. It is an* internal consistency requirement *derivable from the suite's own mathematics: the gauge rank and spacetime dimension are both* outputs *of the framework at a given* K_0*, and they must be equal for the framework's outputs to form a self-consistent whole.*

Proof. The gauge rank $r(K_0)$ is computed in Paper AV from the interface coupling constraint, which is derived from the finite kernel. The spacetime dimension $\tau(K_0)$ is computed in Paper BJ from the dimensional selection theorem, also derived from the finite kernel. Neither computation takes the other as input.

Their equality at $K_0 = 7$ is therefore a non-trivial coincidence within the framework's own machinery. It is the statement that the framework is *self-consistent at* $K_0 = 7$: the gauge structure and the spacetime structure, independently derived from the same finite combinatorial input, are commensurate. □

Remark 4.2 (The question is answered, but not closed). Theorem 3.1 shows that $K_0 = 7$ is *the unique odd prime that passes the internal consistency test*. This answers the question "why $K_0 = 7$?" at the level of the suite's mathematical self-consistency.

What this does *not* prove is that the NFC model with $K_0 = 7$ is the only consistent finite relational framework, or that K_0 must be prime, or that the self-consistency condition (F1)–(F3) is the *only* relevant condition. Those would require:

(a) a proof that K_0 must be an odd prime (from the kernel axioms);

(b) a proof that conditions (F1)–(F3) are exhaustive (no further consistency conditions could eliminate $K_0 = 7$);

(c) a classification of all consistent finite relational frameworks.

These are natural next questions. The present paper answers the *motivational* question: the suite identifies $K_0 = 7$ not by fiat, but because it is the unique value for which the framework's own structural outputs are mutually consistent.

5 The coincidence $r = \tau = 4$ in the Standard Model

Remark 5.1 (Known physics context). The coincidence $r = \tau = 4$ has a known physical correlate: the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ has rank $4 = 2 + 1 + 1$, and spacetime has $4 = 3+1$ dimensions. This coincidence has been noted in the physics literature (e.g. in discussions of gauge coupling unification and anomaly cancellation) but has not, prior to the present framework, been derived from a common combinatorial source.

The NFC framework provides the first *derivation* of this coincidence from a single underlying structure ($K_0 = 7$): the rank 4 of the gauge group and the dimension 4 of spacetime both arise from the same collar alphabet, and their equality follows from the self-consistency of the $7 = 3+2+1+1$ decomposition with the $9 \to 4$ dimensional selection.

6 Status ledger

Label	Claim	Status	Conditions
SC-1	$r(p) = \tau(p)$ is an internal consistency condition	Unconditional	Definition
SC-2	$r(3) = 2 \neq 3 = \tau(3)$: $p = 3$ fails	Unconditional	Computation
SC-3	$r(5) = 3 = \tau(5)$ but no $\mathfrak{u}(1)$: $p = 5$ fails (F3)	Unconditional	Computation
SC-4	$r(7) = 4 = \tau(7) = 4$ with full SM algebra: $p = 7$ passes	Unconditional	Papers AV, BJ
SC-5	$r(p) > \tau(p)$ for $p \geq 11$	Unconditional	Rank growth bound
SC-6	$K_0 = 7$ is the unique odd prime satisfying (F1)–(F3)	Unconditional	SC-2,3,4,5
SC-7	Self-reference: $r(K_0) = \tau(K_0)$ is not an external postulate	Unconditional	SC-6 + framework structure

All entries are Unconditional. The self-consistency argument does not depend on Phase-A or SA2; it is a statement about the structure of the suite's derivations.

7 Conclusion

The question "why $K_0 = 7$?" has a structural answer within the suite's own mathematics. $K_0 = 7$ is not chosen because the Standard Model happens to have gauge group $SU(3) \times SU(2) \times U(1)$ and spacetime has four dimensions. Rather, $K_0 = 7$ is the unique odd prime for which the suite's two independently derived outputs — the gauge rank $r(7) = 4$ and the spacetime dimension $\tau(7) = 4$ — are equal.

The self-consistency condition $r(K_0) = \tau(K_0)$ is derivable from the framework: both outputs are computed from K_0 alone, and their equality is a non-trivial constraint that only $K_0 = 7$ (among odd primes) satisfies while also delivering the correct gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

The suite therefore selects $K_0 = 7$ by self-reference: the NFC framework is self-consistent at exactly one physically viable collar alphabet size, and that size is 7.

The remaining open questions are: (a) why must K_0 be an odd prime (a kernel-axiom question), (b) whether conditions (F1)–(F3) are exhaustive, and (c) whether any other finite relational framework (with different kernel axioms) produces an equally self-consistent picture at a different parameter value. These are natural extensions of the program, not gaps in its current state.

Issued: March 21, 2026. Project Lead. This paper addresses the question left open by Paper I Revision 7. No further proof obligations arise from this analysis.